PROJECT SUMMARY: SOLAR POTENTIAL CALCULATOR FOR ROOFTOPS

TEAM:

5

WRITTEN BY:

Albert Nii Okai Quaye Acromond

Kofi Asante Gyamebi

Kingsley Wunzooya Nahyi

Gerard Ledoux Mendjemo Ngangom

Williewise Oppong

ASHESI UNIVERSITY

(COHORT C, CS213)

OBJECT-ORIENTED PROGRAMMING

FACULTY:

Dr. Daniel Ado

Solar Potential Calculator for Rooftops

This project is a Java-based Object-Oriented Programming project that seeks to develop a system to calculate the potential energy of rooftops of different calibers and determine which household appliances can be powered by these rooftops based on the calculations. The system utilizes Object Oriented Programming employing classes, inheritance, subclasses, and concepts such as encapsulation, polymorphism, and abstraction to accurately define these calculations and provide recommendations based on this framework. The implementation should allow for the real-time compilation of illuminance data from light sensors, such as smartphone ambient light sensors through specialized data collector apps installed on the device. This allows to determine the solar potential of rooftops of different sizes mounted at various angles of incidence to any light source, including sunlight. A practical application of this tool will be to determine the power output of solar panels that can be installed on rooftops adhering to the International Electrotechnical Commission (IEC) standard of having a direct normal irradiance (DNI) measurement of more than 600 W/m^2 (International Electrotechnical Commission [IEC], 2016), power output which could then be used to decide what household appliances could allow for the usage of such solar panels. The solar irradiance at normal incidence (DNI) to a rooftop's surface can be calculated using its linear relationship with illuminance, demonstrated by José Luis Di Laccio et al. (2024) in their research on using smartphone light sensors as innovative tools for solar irradiance measurements. By the end of this project, we intend to find ways to promote renewable energy adoption by providing homeowners with practical insights into their rooftop's energy generation potential and its application to everyday household needs.

References

Di Laccio, J. L., Ferrero, A., Muscas, C., & Serpi, A. (2019). *Using smartphone ambient light sensors for solar irradiance measurements: A feasibility study. Measurement*, 136, 453–460. https://arxiv.org/abs/2308.16158

International Electrotechnical Commission. (2016). *Photovoltaic system performance – Part 3: Energy evaluation method* (IEC TS 61724-3:2016). https://webstore.iec.ch/en/publication/25466